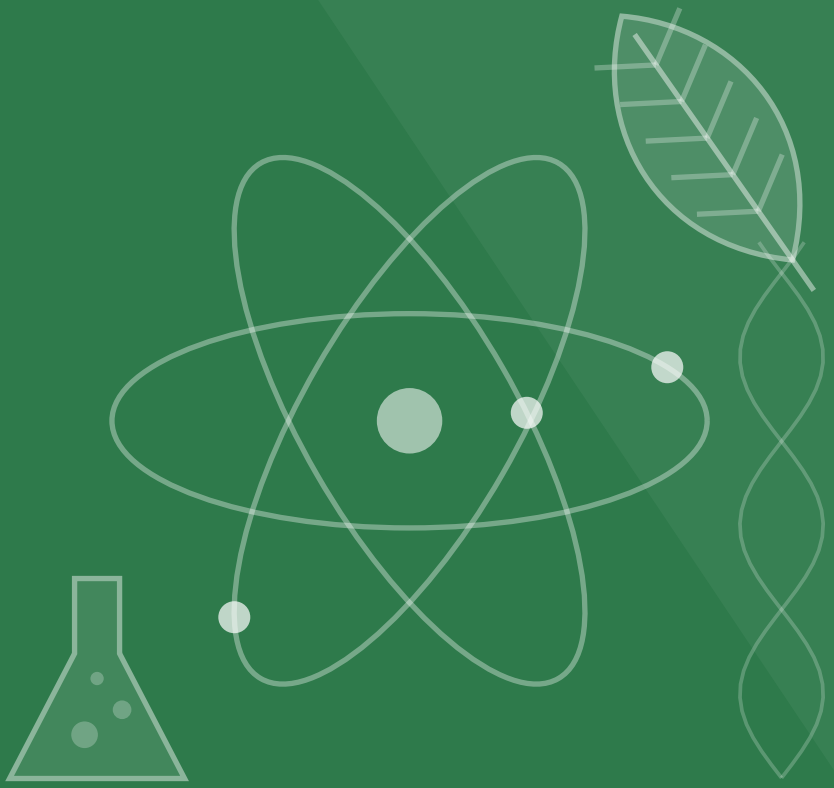
VENVLEARNING

STUDENT WORKBOOK



Science

GRADE 4

Daily practice, weekly checks, and unit tests

2 units

A VenvLearning family homeschool curriculum

GRADE 4 · SCIENCE

VENVLEARNING

Grade 4
Science

STUDENT WORKBOOK

Unit 1 The Human Body

Unit 2 Energy, Heat, and Light

A VenvLearning family homeschool curriculum

How to use this workbook

Every lesson in your textbook has matching practice pages here. Show your thinking in the work space. When a problem has an answer line, write your final answer there.

- Warm-Up reviews something you already know.
- Practice uses today's lesson.
- Think Deeper problems ask you to explain or find a mistake.
- Exit Ticket shows your teacher what you learned today.

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UNIT 1

The Human Body

Thinking like a scientist; cells, tissues, and organs; and the body systems that keep you alive.

Your practice pages for Unit 1. Each lesson matches the same lesson number in your textbook.

UNIT VOCABULARY

observation	Information gathered with senses or tools.
inference	A conclusion based on observations.
fair test	Only one variable is changed.
cell	The smallest living unit.
organ system	Organs that work together.
nutrients	Substances in food the body uses.

LESSON 1.1 · PRACTICE
Week 1 · Lesson 1 of 2

Unit 1: The Human Body

Lesson 1.1: Thinking Like a Scientist

Name _____ Date _____

TODAY'S GOAL

I can tell the difference between an observation and an inference, and ask a question that can be investigated.

Standards: Science and Engineering Practices 1, 3

WHAT TO DO

- [ ] Start with Part A Warm-Up.
- [ ] Read Textbook Lesson 1.1 with your teacher.
- [ ] Then do the rest of these pages in order: Part B Lab Sheet, Part C Check Your Understanding, Part D Practice, Part E Exit Ticket.
- [ ] Answer in complete sentences unless a line asks for one word.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Warm-Up

1. Name two of your senses.

2. Name one tool that helps scientists observe.

PART B Lab Sheet

Directions: Gather what you need. Do each step and check it off. Record as you go.

INVESTIGATE · MYSTERY BAG

What can you figure out about an object you can't see?

You need: a paper bag with a hidden object (chosen by your teacher); paper and a pencil.

1. Without looking, shake the bag and listen. Feel the object through the paper. ☐
2. List at least four observations (sound, weight, shape, texture). ☐
3. Write one inference about what the object is, and the observations that support it. ☐
4. Open the bag. Was your inference correct? ☐

3 Record your results in the table.

Observation (what I sensed)	Inference (what I think it means)

4 Open the bag. What was the object? Was your inference correct?

PART C Check Your Understanding

Directions: Write observation or inference for each sentence, or do what the item asks.

5 Observation or inference: “The leaf is green and has jagged edges.”

6 Observation or inference: “The dog is hungry because it is barking at its bowl.”

7 Make this question testable: “Why do plants like light?”

PART D Practice

8 “The water is 20°C.” Observation or inference?

9 “The ice melted because the room is warm.” Observation or inference?

10 “The rock feels rough.” Observation or inference?

11 “A bird built this nest.” Observation or inference?

12 Look out a window. Write two observations and one inference.

13 Write a testable question about something in your home.

PART E Exit Ticket

14 In your own words, what is the difference between an observation and an inference?

LESSON 1.2 · PRACTICE
Week 1 · Lesson 2 of 2

Unit 1: The Human Body

Lesson 1.2: Fair Tests and Evidence

Name _____ Date _____

TODAY'S GOAL

I can plan a fair test and use evidence to support a claim.

Standards: Science and Engineering Practices 3, 4, 7

WHAT TO DO

- [ ] Start with Part A Warm-Up.
- [ ] Read Textbook Lesson 1.2 with your teacher.
- [ ] Then do the rest of these pages in order: Part B Lab Sheet, Part C Check Your Understanding, Part D Practice, Part E Exit Ticket.
- [ ] Answer in complete sentences unless a line asks for one word.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Warm-Up

1. What does a fair test change?

2. What do you call data that supports a claim?

PART B

Lab Sheet

Directions: Gather what you need. Do each step and check it off. Record as you go.

INVESTIGATE · THE PAPER TOWEL TEST

Which paper towel absorbs the most water?

You need: 2 or 3 kinds of paper towel cut to the same size; measuring cup; water; 3 cups.

1. Pour 100 mL of water into each cup. ☐
2. Dip one towel square in each cup for 10 seconds. ☐
3. Lift the towel and let it drip back into the cup for 10 seconds. ☐
4. Measure the water left in each cup. ☐

3 Record your results in the table.

Towel brand	Water left in cup (mL)	Water absorbed (100 – left)

4 Write a CER: Which paper towel absorbs the most water?

Claim My answer to the question	
Evidence Data from my investigation	
Reasoning Why my evidence supports my claim	

PART C Check Your Understanding

Directions: Answer each question.

5 Why must the towel squares be the same size?

6 What is the claim in CER?

7 A student tests plant growth but gives one plant more water AND more light. What is wrong?

PART D Practice

8 You want to know if a ball bounces higher on carpet or wood. What will you change? What will you keep the same? What will you measure?

9 Brand A towel left 40 mL; Brand B left 65 mL; Brand C left 52 mL (from 100 mL). Which absorbed the most?

10 Write a CER for the towel data above.

PART E Exit Ticket

11 Why do scientists keep all variables the same except one?

WEEKLY CHECK
After Lesson 1.2

About 15 minutes

Weekly Check 1: Thinking Like a Scientist

Name _____ Date _____

DIRECTIONS

- [ ] Work on your own. Don't use your textbook unless your teacher says you may.
- [ ] Read each passage and question carefully. Answer in complete sentences.
- [ ] When you finish, check that every question has an answer.

Show what you know

Directions: Write observation or inference for each sentence, or do what the item asks.

1. Observation or inference: “The sky is cloudy.”

2. Observation or inference: “It will probably rain.”

3. Make this testable: “Why are some balls bouncy?”

4. A student tests whether salt water or fresh water freezes faster, but puts one cup in the freezer and one in the refrigerator. What is wrong?

5. What does the E in CER stand for?

LESSON 1.3 · PRACTICE
Week 2 · Lesson 1 of 2

Unit 1: The Human Body

Lesson 1.3: Cells, Tissues, Organs, and Systems

Name _____ Date _____

TODAY'S GOAL

I can describe how the body is organized from cells to organ systems.

Standards: 4-LS1-1

WHAT TO DO

- [ ] Start with Part A Warm-Up.
- [ ] Read Textbook Lesson 1.3 with your teacher.
- [ ] Then do the rest of these pages in order: Part B Lab Sheet, Part C Check Your Understanding, Part D Practice, Part E Exit Ticket.
- [ ] Answer in complete sentences unless a line asks for one word.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Warm-Up

1. A group of similar cells is a...

2. The heart is an...

PART B Lab Sheet

Directions: Gather what you need. Do each step and check it off. Record as you go.

INVESTIGATE · BUILD THE LEVELS

How can a model show levels of organization?

You need: building blocks or LEGO bricks; sticky notes.

1. Let one brick stand for one cell. ☐
2. Join several matching bricks into a "tissue." ☐
3. Combine different "tissues" into an "organ." ☐
4. Put several organs together into a "system." Label each level with a sticky note. ☐

3 Sketch your brick model. Label each level: cell, tissue, organ, organ system.

PART C Check Your Understanding

Directions: Answer each question.

4 What is the smallest living unit of the body?

5 Put in order: organ, cell, organ system, tissue

6 Name one organ.

PART D Practice

Directions: Answer each question about how living things are organized.

7 Smallest level of organization

8 Organs working together form an...

9 Why are nerve cells long and thin?

10 Is skin an organ?

11 Use the brick-and-house comparison to explain levels of organization.

PART E Exit Ticket

12 Draw and label the four levels, from cell to organ system.

LESSON 1.4 · PRACTICE
Week 2 · Lesson 2 of 2
Unit 1: The Human Body

Lesson 1.4: Body Systems Work Together

Name _____ Date _____

TODAY'S GOAL

I can name major body systems, describe each one's main job, and explain how they work together.

Standards: 4-LS1-1

WHAT TO DO

- [ ] Start with Part A Warm-Up.
- [ ] Read Textbook Lesson 1.4 with your teacher.
- [ ] Then do the rest of these pages in order: Part B Lab Sheet, Part C Check Your Understanding, Part D Match the System, Part E Apply, Part F Exit Ticket.
- [ ] Answer in complete sentences unless a line asks for one word.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Warm-Up

Directions: Write the body system that fills each blank.

1. The brain belongs to the ___ system.

2. Bones belong to the ___ system.

PART B Lab Sheet

Directions: Gather what you need. Do each step and check it off. Record as you go.

INVESTIGATE · SYSTEMS RELAY

Which systems work when you do an everyday task?

You need: index cards labeled with the six systems; a timer.

1. Choose an action: eating an apple, running up stairs, or reading a book aloud. ☐
2. Do the action (safely!). ☐
3. Hold up every system card you think was working, and explain why. ☐

3 Record your results in the table.

My action	Systems I used	Why each system was needed

PART C Check Your Understanding

Directions: Answer each question.

4 Which system carries oxygen to cells?

5 Which system breaks down food?

6 Which systems help you jump?

PART D Match the System

Directions: Write the body system each organ belongs to.

7 lungs

8 heart

9 stomach

10 spinal cord

11 skull

12 biceps

PART E Apply

13 Choose an activity (swimming, playing piano, eating lunch). Explain how three systems work together.

PART F Exit Ticket

14 Why do you breathe faster when you run?

WEEKLY CHECK
After Lesson 1.4

About 15 minutes

Weekly Check 2: Body Systems

Name _____ Date _____

DIRECTIONS

- [ ] Work on your own. Don't use your textbook unless your teacher says you may.
- [ ] Read each passage and question carefully. Answer in complete sentences.
- [ ] When you finish, check that every question has an answer.

Show what you know

1. Put in order: organ, cell, tissue, organ system

2. Which system includes the brain?

3. Which system breaks down food?

4. Which system moves blood?

5. Name two systems that work together when you climb stairs and explain how.

LESSON 1.5 · PRACTICE
Week 3 · Lesson 1 of 2

Unit 1: The Human Body

Lesson 1.5: The Digestive System

Name _____ Date _____

TODAY'S GOAL

I can describe the path of food through the digestive system and explain what each organ does.

Standards: 4-LS1-1

WHAT TO DO

- [ ] Start with Part A Warm-Up.
- [ ] Read Textbook Lesson 1.5 with your teacher.
- [ ] Then do the rest of these pages in order: Part B Lab Sheet, Part C Check Your Understanding, Part D Practice, Part E Exit Ticket.
- [ ] Answer in complete sentences unless a line asks for one word.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Warm-Up

1. What do teeth do to food?

2. What are nutrients?

PART B Lab Sheet

Directions: Gather what you need. Do each step and check it off. Record as you go.

INVESTIGATE · STOMACH IN A BAG

How does the stomach help break down food?

You need: zip-top plastic bag; 4 crackers; 1/2 cup water; a few drops of vinegar (to stand for acid).

1. Put the crackers in the bag. Observe them. ☐
2. Add the water and vinegar and seal the bag tightly. ☐
3. Gently squeeze and mash the bag for 2 minutes, the way stomach muscles churn. ☐
4. Observe the crackers again. ☐

Safety: Do not eat or drink anything used in an investigation.

3 Record your results in the table.

	Before squeezing	After squeezing
How the crackers look		
How the crackers feel		

PART C Check Your Understanding

Directions: Answer each question.

4 Where do most nutrients enter the blood?

5 What does the large intestine absorb?

6 What did squeezing the bag model?

PART D Practice

Directions: Do what each item asks. For mechanical or chemical, write one of those words. For questions with choices, circle the letter of the best answer.

7 Put in order: stomach, mouth, large intestine, esophagus, small intestine

8 Mechanical or chemical: teeth chewing

9 Mechanical or chemical: stomach acid

10 Which organ is a long, coiled tube where nutrients enter the blood?

A esophagus
B stomach
C small intestine
D large intestine

11 Explain why food must be broken down before the body can use it.

PART E Exit Ticket

12 Describe what happened to the crackers in the bag and what it shows about digestion.

LESSON 1.6 · PRACTICE
Week 3 · Lesson 2 of 2
Unit 1: The Human Body

Lesson 1.6: The Respiratory System

Name _____ Date _____

TODAY'S GOAL

I can explain how the respiratory system brings oxygen into the body and removes carbon dioxide.

Standards: 4-LS1-1

WHAT TO DO

- [ ] Start with Part A Warm-Up.
- [ ] Read Textbook Lesson 1.6 with your teacher.
- [ ] Then do the rest of these pages in order: Part B Lab Sheet, Part C Check Your Understanding, Part D Practice, Part E Exit Ticket.
- [ ] Answer in complete sentences unless a line asks for one word.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Warm-Up

1. The gas you breathe out is...

2. The windpipe is also called the...

PART B

Lab Sheet

Directions: Gather what you need. Do each step and check it off. Record as you go.

INVESTIGATE · BREATHING RATE

How does exercise change how fast you breathe?

You need: a timer; a partner.

1. Sit quietly for 1 minute. Count your breaths for 30 seconds and multiply by 2. ☐
2. Do jumping jacks for 1 minute. ☐
3. Right away, count your breaths for 30 seconds and multiply by 2. ☐
4. Rest 3 minutes and count again. ☐

3 Record your results in the table.

When I counted	Breaths in 30 seconds	Breaths per minute (× 2)
At rest		
Right after exercise		
After 3 minutes of rest		

4 Make a bar graph of your breathing data: at rest, after exercise, after resting.

PART C Check Your Understanding

Directions: Answer each question.

5 What gas do your cells need?

6 What muscle helps you breathe?

7 Why does breathing speed up during exercise?

PART D Practice

8 Order: air sacs, nose, windpipe, bronchial tubes

9 Where does oxygen pass into the blood?

10 Maya breathed 18 times per minute at rest and 36 after exercise. How many more?

11 Explain how the respiratory and circulatory systems work together.

PART E Exit Ticket

12 What happens when the diaphragm moves down?

WEEKLY CHECK
After Lesson 1.6

About 15 minutes

Weekly Check 3: Digestion and Breathing

Name _____ Date _____

DIRECTIONS

- [ ] Work on your own. Don't use your textbook unless your teacher says you may.
- [ ] Read each passage and question carefully. Answer in complete sentences.
- [ ] When you finish, check that every question has an answer.

Show what you know

Directions: Answer each question. For questions with choices, circle the letter of the best answer.

1. Where do most nutrients enter the blood?

2. What does the large intestine absorb?

3. What gas do you breathe out?

4. Where are gases exchanged in the lungs?

5. Which is mechanical digestion?

   A stomach acid
   B chewing
   C saliva chemicals

6. Ben breathes 16 times a minute at rest and 34 after running. Explain why.

LESSON 1.7 · PRACTICE
Week 4 · Lesson 1 of 2

Unit 1: The Human Body

Lesson 1.7: The Heart and Blood Vessels

Name _____ Date _____

TODAY'S GOAL

I can describe how the heart and blood vessels move blood through the body, and measure my pulse.

Standards: 4-LS1-1

WHAT TO DO

- [ ] Start with Part A Warm-Up.
- [ ] Read Textbook Lesson 1.7 with your teacher.
- [ ] Then do the rest of these pages in order: Part B Lab Sheet, Part C Check Your Understanding, Part D Practice, Part E Exit Ticket.
- [ ] Answer in complete sentences unless a line asks for one word.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Warm-Up

Directions: Finish each sentence.

1. The heart is about the size of your...

2. Veins carry blood...

PART B Lab Sheet

Directions: Gather what you need. Do each step and check it off. Record as you go.

INVESTIGATE · TAKE YOUR PULSE

How does activity change heart rate?

You need: a timer.

1. Place two fingers (not your thumb) on the side of your neck or the inside of your wrist. ☐
2. Count the beats for 15 seconds. Multiply by 4 to find beats per minute. ☐
3. Run in place for 1 minute, then measure again. ☐
4. Rest 3 minutes and measure again. ☐

3 Record your results in the table.

When I counted	Beats in 15 seconds	Beats per minute (× 4)
At rest		
After running in place		
After 3 minutes of rest		

4 Compare with your breathing data from Lesson 1.6. What happened to both during exercise?

PART C Check Your Understanding

Directions: Answer each question.

5 Which vessels carry blood away from the heart?

6 Where do oxygen and nutrients move into cells?

7 15-second count is 22 beats. Beats per minute?

PART D Practice

Directions: Multiply each 15-second pulse count by 4 to find beats per minute, or answer the question. For questions with choices, circle the letter of the best answer.

8 Pulse count of 20 in 15 seconds = ___ beats per minute

9 Pulse count of 30 in 15 seconds = ___ beats per minute

10 Why does heart rate go up during exercise?

11 Which vessel is the tiniest?

A artery
B vein
C capillary

12 Trace a drop of blood from the heart to the lungs and back to the body.

PART E Exit Ticket

13 Why do the respiratory and circulatory systems both speed up during exercise?

LESSON 1.8 · PRACTICE
Week 4 · Lesson 2 of 2

Unit 1: The Human Body

Lesson 1.8: Bones, Muscles, and Healthy Choices

Name _____ Date _____

TODAY'S GOAL

I can explain how bones and muscles work together and describe choices that keep body systems healthy.

Standards: 4-LS1-1

WHAT TO DO

- [ ] Start with Part A Warm-Up.
- [ ] Read Textbook Lesson 1.8 with your teacher.
- [ ] Then do the rest of these pages in order: Part B Lab Sheet, Part C Check Your Understanding, Part D Practice, Part E Healthy Choices, Part F Exit Ticket.
- [ ] Answer in complete sentences unless a line asks for one word.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Warm-Up

1. What do the ribs protect?

2. A place where bones meet is a...

PART B Lab Sheet

Directions: Gather what you need. Do each step and check it off. Record as you go.

INVESTIGATE · MUSCLE PARTNERS

How do muscles work in pairs?

You need: your own arm.

1. Place your hand on your biceps. Slowly bend your arm. What does the muscle do? ☐
2. Move your hand to your triceps. Straighten your arm. What does that muscle do? ☐

3 Sketch your upper arm. Label the muscle that pulls to bend it (biceps) and the one that pulls to straighten it (triceps).

PART C Check Your Understanding

Directions: Answer each question.

4 How many bones in an adult skeleton?

5 What connects muscle to bone?

6 Why do muscles work in pairs?

PART D Practice

Directions: Write the name of each muscle or body part.

7 Muscle that bends the arm

8 Muscle that straightens the arm

9 Connects bone to bone

10 The skull protects the...

PART E Healthy Choices

11 Write a one-day plan with three healthy choices. Name the system each one helps.

PART F Exit Ticket

12 Describe how the skeletal and muscular systems work together to help you walk.

UNIT REVIEW
Study before the unit test

About 40 minutes

Unit 1 Review

Name _____ Date _____

DIRECTIONS

- [ ] Work on your own. Don't use your textbook unless your teacher says you may.
- [ ] Read each passage and question carefully. Answer in complete sentences.
- [ ] When you finish, check that every question has an answer.

PART A Scientists

Directions: Answer each item.

1. Observation or inference: “The cat is sleepy.”

2. In a fair test, how many variables do you change on purpose?

PART B Body systems

Directions: Write the name of each body part.

3. Smallest unit of life

4. Carries blood back to the heart

5. Tiny vessels where materials pass to cells

6. Muscle that bends the arm

7. Tube from mouth to stomach

8. Muscle under the lungs

PART C Explain

9 How do the digestive, respiratory, and circulatory systems work together to give a muscle cell energy?

UNIT TEST
At the end of the unit

About 45 minutes

Unit 1 Test

Name _____ Date _____

DIRECTIONS

- [ ] Work on your own. Don't use your textbook unless your teacher says you may.
- [ ] This test has 21 points.
- [ ] Read each passage and question carefully. Answer in complete sentences.
- [ ] When you finish, check that every question has an answer.

PART A Part A · Scientists (1 point each)

Directions: Write observation or inference for each sentence, or do what the item asks.

1. Observation or inference: “The soup is steaming.”

2. Observation or inference: “The soup is too hot to eat.”

3. Plan a fair test: Does a ball roll farther on carpet or on tile? Name what you change, keep the same, and measure.

PART B Part B · Body systems (1 point each)

4. Which list is in order from smallest to largest?

   A organ, tissue, cell, system
   B cell, tissue, organ, system
   C tissue, cell, system, organ

5. Which system includes the lungs?

6 Which system includes the brain and nerves?

7 Where do most nutrients pass into the blood?

8 What gas do cells need?

9 Which blood vessels carry blood away from the heart?

10 What connects a muscle to a bone?

11 Why do muscles work in pairs?

12 Which is chemical digestion?

A chewing
B stomach churning
C stomach acid breaking down food

PART C Part C · Data (1 point each)

13 A pulse of 24 beats in 15 seconds is how many beats per minute?

14 Resting breaths: 18/min; after exercise: 38/min. What changed and by how much?

15 Write a CER: Does exercise increase heart rate? Use this data: resting 80, after running 128 beats per minute.

PART D Part D · Explain (3 points)

16 Choose an activity. Explain how at least three body systems work together during it.

UNIT 2

Energy, Heat, and Light

Forms of energy, heat and temperature, conduction, convection, and radiation, conductors and insulators, how light travels, reflection, refraction, and color.

Your practice pages for Unit 2. Each lesson matches the same lesson number in your textbook.

UNIT VOCABULARY

energy	The ability to make something move or change.
heat	Energy flowing from warmer to cooler.
conductor	Lets heat move easily.
insulator	Slows heat.
reflect	Bounce off.
refraction	The bending of light.

LESSON 2.1 · PRACTICE
Week 1 · Lesson 1 of 2

Unit 2: Energy, Heat, and Light

Lesson 2.1: What Is Energy?

Name _____ Date _____

TODAY'S GOAL

I can describe forms of energy and explain that faster objects have more energy.

Standards: 4-PS3-1, 4-PS3-2

WHAT TO DO

- [ ] Start with Part A Warm-Up.
- [ ] Read Textbook Lesson 2.1 with your teacher.
- [ ] Then do the rest of these pages in order: Part B Lab Sheet, Part C Check Your Understanding, Part D Practice, Part E Exit Ticket.
- [ ] Answer in complete sentences unless a line asks for one word.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Warm-Up

Directions: Write the word for each description.

1. The ability to make something move or change

_____

2. To move from one object to another

_____

PART B Lab Sheet

Directions: Gather what you need. Do each step and check it off. Record as you go.

INVESTIGATE · RAMP AND CUP

Does a faster marble have more energy?

You need: a ruler with a groove (or a cardboard ramp); a stack of books; a marble; a paper cup with a doorway cut out; a measuring tape.

1. Set the ramp on one book and place the cup at the bottom. ☐
2. Let the marble go (don't push it) and measure how far the cup moves. ☐
3. Repeat with two books, then three books. ☐
4. Do three trials at each height and record the distances. ☐

3 Record how far the cup moved each time.

Ramp height	Trial 1 (cm)	Trial 2 (cm)	Trial 3 (cm)
1 book			
2 books			
3 books			

4 Use your Ramp and Cup data. What happened as the ramp got higher? What does that tell you about speed and energy?

PART C Check Your Understanding

Directions: Answer each question.

5 Name three forms of energy.

6 Which has more motion energy: a slow bike or a fast bike?

7 What energy changes happen in a flashlight?

PART D Practice

Directions: Write the form of energy you notice in each example.

8 a ringing bell

9 a toaster's glowing wires

10 a kicked soccer ball

11 a phone charger

PART E Exit Ticket

Directions: Answer in complete sentences.

12 Explain why a fast-moving ball hurts more when it hits you than a slow one.

LESSON 2.2 · PRACTICE
Week 1 · Lesson 2 of 2

Unit 2: Energy, Heat, and Light

Lesson 2.2: Heat and Temperature

Name _____ Date _____

TODAY'S GOAL

I can measure temperature and explain that heat flows from warmer objects to cooler ones.

Standards: 4-PS3-2

WHAT TO DO

- [ ] Start with Part A Warm-Up.
- [ ] Read Textbook Lesson 2.2 with your teacher.
- [ ] Then do the rest of these pages in order: Part B Lab Sheet, Part C Check Your Understanding, Part D Practice, Part E Exit Ticket.
- [ ] Answer in complete sentences unless a line asks for one word.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Warm-Up

Directions: Write the word or number for each description.

1. How hot or cold something is

2. Water boils at… (°C)

PART B

Lab Sheet

Directions: Gather what you need. Do each step and check it off. Record as you go.

INVESTIGATE · COOLING CUPS

How does the temperature of warm water change over time?

You need: two cups; warm tap water and cold tap water; a thermometer; a timer.

1. Fill one cup with warm tap water and one with cold tap water. ☐
2. Measure and record each temperature. ☐
3. Measure both cups again every 5 minutes for 20 minutes. ☐
4. Measure the room's air temperature too. ☐

Safety: Use warm tap water, not hot or boiling water.

3 Record your results in the table.

Time	Warm cup (°F)	Cold cup (°F)
Start		
5 minutes		
10 minutes		
15 minutes		
20 minutes		

4 Room air temperature (°F):

5 Use your Cooling Cups data. What happened to each cup's temperature? Where did each cup end up compared with the room?

PART C Check Your Understanding

Directions: Answer each question.

6 What tool measures temperature?

7 At what temperature does water freeze in °F?

8 You hold a snowball. Which way does heat flow?

PART D Practice

Directions: Tell which way heat flows in each example.

9 a hot pan cooling on the stove

10 a cold drink on a summer day

11 your feet on a cold tile floor

12 an ice cube in your hand

PART E Exit Ticket

Directions: Answer in complete sentences.

13 Explain why hot cocoa gets cold if you leave it on the table.

WEEKLY CHECK
After Lesson 2.2

About 15 minutes

Weekly Check 1: Energy and Heat

Name _____ Date _____

DIRECTIONS

- [ ] Work on your own. Don't use your textbook unless your teacher says you may.
- [ ] Read each passage and question carefully. Answer in complete sentences.
- [ ] When you finish, check that every question has an answer.

Show what you know

Directions: Answer each question about energy and heat.

1. The ability to make something move or change

2. Which has more motion energy: a fast car or a slow car?

3. A tool that measures temperature

4. Water freezes at... (°F)

5. You hold an ice cube. Which way does heat flow?

6. Name two forms of energy and give an example of each.

LESSON 2.3 · PRACTICE
Week 2 · Lesson 1 of 2

Unit 2: Energy, Heat, and Light

Lesson 2.3: How Heat Moves

Name _____ Date _____

TODAY'S GOAL

I can describe three ways heat moves: conduction, convection, and radiation.

Standards: 4-PS3-2

WHAT TO DO

- [ ] Start with Part A Warm-Up.
- [ ] Read Textbook Lesson 2.3 with your teacher.
- [ ] Then do the rest of these pages in order: Part B Lab Sheet, Part C Check Your Understanding, Part D Sort, Part E Exit Ticket.
- [ ] Answer in complete sentences unless a line asks for one word.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Warm-Up

Directions: Write the word for each description.

1. Heat moving through touching objects

_____

2. Heat traveling as waves

_____

PART B Lab Sheet

Directions: Gather what you need. Do each step and check it off. Record as you go.

INVESTIGATE · FOOD COLORING CURRENTS

Can you see convection in water?

You need: a clear glass dish or jar of room-temperature water; a small cup of warm water with red food coloring; an ice cube made with blue food coloring.

1. Let the water in the dish sit until it is still. ☐
2. Gently set the blue ice cube at one end. ☐
3. Carefully pour a little red warm water in at the other end, near the bottom. ☐
4. Watch from the side for 5 minutes. Where does each color go? ☐

Safety: Use warm tap water only.

3 Draw a side view of your container. Use one color of arrows for warm water and another for cold water. Make a key.

PART C Check Your Understanding

Directions: Answer each question.

4 How does the sun's heat reach Earth?

5 A pan gets hot on a burner. Which way does the heat move?

6 Why is the top floor of a house often warmer?

PART D Sort

Directions: Write conduction, convection, or radiation for each example.

7 warming your hands near a fireplace

8 a metal slide getting hot in the sun and burning your legs

9 warm air from a vent rising to the ceiling

10 an ice pack cooling a bump

11 soup bubbling and churning in a pot

12 sunlight warming a car seat

PART E Exit Ticket

Directions: Answer in complete sentences.

13 Describe the convection you saw in the Food Coloring Currents investigation.

LESSON 2.4 · PRACTICE
Week 2 · Lesson 2 of 2

Unit 2: Energy, Heat, and Light

Lesson 2.4: Conductors and Insulators

Name _____ Date _____

TODAY'S GOAL

I can test materials and design a way to slow the flow of heat.

Standards: 4-PS3-2, 4-PS3-4, 3-5-ETS1-1

WHAT TO DO

- [ ] Start with Part A Warm-Up.
- [ ] Read Textbook Lesson 2.4 with your teacher.
- [ ] Then do the rest of these pages in order: Part B Lab Sheet, Part C Check Your Understanding, Part D Sort, Part E Exit Ticket.
- [ ] Answer in complete sentences unless a line asks for one word.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Warm-Up

Directions: Write the word for each description.

1. A material that slows heat

2. What a design must do to succeed

PART B

Lab Sheet

Directions: Gather what you need. Do each step and check it off. Record as you go.

INVESTIGATE · ICE CUBE KEEPER

Which design keeps an ice cube frozen longest?

You need: three ice cubes of the same size; three small cups; insulating materials: cloth, cotton balls, foam, newspaper, foil; a timer.

1. Leave one ice cube in a plain cup (your control). ☐
2. Design and build two holders using different insulators. ☐
3. Start the timer and check every 10 minutes. ☐
4. Record when each ice cube is completely melted. ☐

3 Record your results in the table.

Holder	Insulator used	Minutes until melted
Plain cup (control)		
Design 1		
Design 2		

4 Sketch your best Ice Cube Keeper and label the materials.

5 How would you improve your design if you tested it again?

6 Which material was the best insulator? Use your data.

PART C Check Your Understanding

Directions: Answer each question.

7 Is copper a conductor or an insulator?

8 Why are oven mitts made of thick cloth?

9 What are the steps of the design process?

PART D Sort

Directions: Write conductor or insulator for each material.

10 a metal spoon

11 a wool sweater

12 a foam cooler

13 an aluminum pan

14 a rubber pot holder

15 an iron skillet

PART E Exit Ticket

Directions: Answer in complete sentences.

16 Why does a cooler keep drinks cold on a hot day?

WEEKLY CHECK
After Lesson 2.4

About 15 minutes

Weekly Check 2: How Heat Moves

Name _____ Date _____

DIRECTIONS

- [ ] Work on your own. Don't use your textbook unless your teacher says you may.
- [ ] Read each passage and question carefully. Answer in complete sentences.
- [ ] When you finish, check that every question has an answer.

Show what you know

Directions: Answer each question about how heat moves.

1. Heat moving through touching objects

2. Heat moving by flowing air or water

3. How the sun's heat reaches Earth

4. Conductor or insulator: a copper pot

5. Conductor or insulator: a foam cooler

6. Why are pot handles often made of plastic or wood?

LESSON 2.5 · PRACTICE
Week 3 · Lesson 1 of 2

Unit 2: Energy, Heat, and Light

Lesson 2.5: Light Travels and How We See

Name _____ Date _____

TODAY'S GOAL

I can explain that light travels in straight lines and that we see objects when light reflects off them into our eyes.

Standards: 4-PS4-2

WHAT TO DO

- [ ] Start with Part A Warm-Up.
- [ ] Read Textbook Lesson 2.5 with your teacher.
- [ ] Then do the rest of these pages in order: Part B Lab Sheet, Part C Check Your Understanding, Part D Exit Ticket.
- [ ] Answer in complete sentences unless a line asks for one word.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Warm-Up

Directions: Write the word for each description.

1. To bounce off a surface

_____

2. Something that gives off its own light

_____

PART B

Lab Sheet

Directions: Gather what you need. Do each step and check it off. Record as you go.

INVESTIGATE · SHADOW LENGTHS

How does the position of a light change a shadow?

You need: a flashlight; a small toy or a pencil standing in clay; a ruler; a dark room.

1. Stand the toy on a table and shine the flashlight from straight above. Measure the shadow. ☐
2. Shine it from a medium angle. Measure the shadow. ☐
3. Shine it from low on the side. Measure the shadow. ☐
4. Keep the flashlight the same distance away each time. ☐

3 Record your results in the table.

Flashlight position	Shadow length (cm)
Straight above	
Medium angle	
Low on the side	

4 Draw a model showing how you see a ball in a sunny yard. Use arrows for the path of light from the sun to the ball to your eye.

5 Use your Shadow Lengths data. How did the light's position change the shadow? Why?

PART C Check Your Understanding

Directions: Answer each question.

6 Name two light sources.

7 Why can't you see a book in a dark closet?

8 When is your shadow longest: morning or noon?

PART D Exit Ticket

Directions: Answer in complete sentences.

9 Explain how you see the words on this page.

LESSON 2.6 · PRACTICE
Week 3 · Lesson 2 of 2

Unit 2: Energy, Heat, and Light

Lesson 2.6: Reflection, Absorption, and Transparency

Name _____ Date _____

TODAY'S GOAL

I can describe how light reflects, passes through, or is absorbed by different materials.

Standards: 4-PS4-2

WHAT TO DO

- [ ] Start with Part A Warm-Up.
- [ ] Read Textbook Lesson 2.6 with your teacher.
- [ ] Then do the rest of these pages in order: Part B Lab Sheet, Part C Check Your Understanding, Part D Sort, Part E Apply, Part F Exit Ticket.
- [ ] Answer in complete sentences unless a line asks for one word.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Warm-Up

Directions: Write the word for each description.

1. Blocks light

2. Lets all light pass through

PART B Lab Sheet

Directions: Gather what you need. Do each step and check it off. Record as you go.

INVESTIGATE · LIGHT DETECTIVE

Which materials let light through?

You need: a flashlight; materials: clear plastic, wax paper, foil, cardboard, tissue paper, a mirror, a glass of water.

1. Predict whether each material is transparent, translucent, or opaque. ☐
2. Shine the flashlight through each material onto a wall. ☐
3. Record what you see on the wall: bright light, dim blurry light, or no light. ☐
4. Shine the light at the mirror and find where the reflection lands. ☐

3 Record your results in the table.

Material	Prediction	Result (bright, dim, or no light)

4 Where did the light land when you shone it at the mirror?

PART C Check Your Understanding

Directions: Answer each question.

5 Is wax paper transparent, translucent, or opaque?

6 What happens to light that hits a mirror?

7 Why does a black shirt feel hotter in the sun?

PART D Sort

Directions: Write transparent, translucent, or opaque for each object.

8 a car window

9 a wooden door

10 a lampshade

11 a cereal box

12 a clear water bottle

13 tissue paper

PART E Apply

Directions: Answer in complete sentences.

14 Which material would you choose for a bathroom window that lets in light but keeps privacy? Why?

PART F Exit Ticket

Directions: Answer in complete sentences.

15 Name the three things light can do when it hits a material.

WEEKLY CHECK
After Lesson 2.6

About 15 minutes

Weekly Check 3: Light

Name _____ Date _____

DIRECTIONS

- [ ] Work on your own. Don't use your textbook unless your teacher says you may.
- [ ] Read each passage and question carefully. Answer in complete sentences.
- [ ] When you finish, check that every question has an answer.

Show what you know

Directions: Answer each question about light.

1. Something that gives off its own light

2. Lets some light through, but blurry

3. Blocks light

4. When is a shadow longest: morning or noon?

5. What bounces light to make an image you can see?

6. Explain how you see a tree outside.

LESSON 2.7 · PRACTICE
Week 4 · Lesson 1 of 2

Unit 2: Energy, Heat, and Light

Lesson 2.7: Refraction and Color

Name _____ Date _____

TODAY'S GOAL

I can explain that light bends when it passes into a new material and that white light is made of many colors.

Standards: 4-PS4-2

WHAT TO DO

- [ ] Start with Part A Warm-Up.
- [ ] Read Textbook Lesson 2.7 with your teacher.
- [ ] Then do the rest of these pages in order: Part B Lab Sheet, Part C Check Your Understanding, Part D Practice, Part E Exit Ticket.
- [ ] Answer in complete sentences unless a line asks for one word.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Warm-Up

Directions: Write the word for each description.

1. The bending of light

_____

2. The band of colors in white light

_____

PART B Lab Sheet

Directions: Gather what you need. Do each step and check it off. Record as you go.

INVESTIGATE · MAKE A SPECTRUM

Can you split white light into colors?

You need: a clear glass of water; a sheet of white paper; a sunny window or a bright flashlight.

1. Set the glass of water on a sunny windowsill. ☐
2. Hold the white paper on the floor or table where light passes through the water. ☐
3. Move the paper until you see colors. ☐
4. Record the order of the colors you see. ☐

3 Color your spectrum in the order you saw the colors.

4 List the colors in order.

PART C Check Your Understanding

Directions: Answer each question.

5 Why does a pencil in water look broken?

6 What colors are in white light?

7 Why does a banana look yellow?

PART D Practice

Directions: Answer each question. Use the lesson.

8 Name a tool that uses a lens.

9 What acts like a prism to make a rainbow?

10 Which color does a green leaf reflect?

11 Which reflects the most colors: a white shirt or a black shirt?

PART E Exit Ticket

Directions: Answer in complete sentences.

12 Explain how a rainbow forms.

LESSON 2.8 · PRACTICE
Week 4 · Lesson 2 of 2

Unit 2: Energy, Heat, and Light

Lesson 2.8: Light Investigation: Claim, Evidence, and Reasoning

Name _____ Date _____

TODAY'S GOAL

I can plan an investigation about light and explain my results with a claim, evidence, and reasoning.

Standards: 4-PS4-2, Science and Engineering Practices 3, 7

WHAT TO DO

- [ ] Start with Part A Warm-Up.
- [ ] Read Textbook Lesson 2.8 with your teacher.
- [ ] Then do the rest of these pages in order: Part B Lab Sheet, Part C Check Your Understanding, Part D Exit Ticket.
- [ ] Answer in complete sentences unless a line asks for one word.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Warm-Up

Directions: Write the word for each description.

1. A statement that answers a question

2. Data that supports a claim

PART B

Lab Sheet

Directions: Gather what you need. Do each step and check it off. Record as you go.

INVESTIGATE · BUILD A PERISCOPE

Can mirrors let you see over a wall?

You need: a long cardboard box (like a juice or milk carton, cleaned); two small mirrors; tape; scissors (with an adult).

1. Cut a square window near the top of one side and near the bottom of the opposite side. ☐

2. Tape one mirror inside the top, tilted to face down the tube; tape the other inside the bottom, tilted to face the bottom window. ☐

3. Look through the bottom window and adjust the mirrors until you see out the top. ☐

4. Hide below a table or couch and test what you can see. ☐

Safety: An adult should help with cutting. Handle mirrors carefully.

3 Draw your periscope from the side. Show both mirrors and draw arrows for the path of light into your eye.

4 Write a CER paragraph: Can a periscope let you see over a wall?

PART C Check Your Understanding

Directions: Answer each question.

5 What are the three parts of a scientific explanation?

6 How many mirrors does a periscope use?

7 What science idea explains how a periscope works?

PART D Exit Ticket

Directions: Answer in complete sentences.

8 Why do you need the reasoning part of CER, not just the evidence?

UNIT REVIEW
Study before the unit test

About 40 minutes

Unit 2 Review

Name _____ Date _____

DIRECTIONS

- [ ] Work on your own. Don't use your textbook unless your teacher says you may.
- [ ] Read each passage and question carefully. Answer in complete sentences.
- [ ] When you finish, check that every question has an answer.

PART A Words

Directions: Write what each word means.

1. temperature

2. conduction

3. insulator

4. reflect

5. refraction

6. opaque

PART B Models

Directions: Draw and label each model.

7. Draw arrows to show how heat moves from a stove burner, through a pan, and into food.

8 Draw a model that shows how you see a book in a lit room.

UNIT TEST
At the end of the unit

About 45 minutes

Unit 2 Test

Name _____ Date _____

DIRECTIONS

- [ ] Work on your own. Don't use your textbook unless your teacher says you may.
- [ ] This test has 16 points.
- [ ] Read each passage and question carefully. Answer in complete sentences.
- [ ] When you finish, check that every question has an answer.

PART A · Part A · Energy and heat (1 point each)

Directions: Answer each question. For questions with choices, circle the letter of the best answer.

1. Which marble has the most motion energy?
   A a still marble
   B a slow marble
   C a fast marble

2. Name two forms of energy.

3. Water boils at... (°C)

4. Heat always flows...
   A from cooler to warmer
   B from warmer to cooler
   C only through metal

PART B Part B · How heat moves (1 point each)

Directions: Answer each question. For questions with choices, circle the letter of the best answer.

5 Warm air rising from a heater is an example of...

6 Feeling heat from a campfire is an example of...

7 A spoon getting hot in soup is an example of...

8 Which is the best insulator?

A an aluminum pan
B a wool mitten
C a copper wire

PART C Part C · Light (1 point each)

Directions: Answer each question. For questions with choices, circle the letter of the best answer.

9 Why can't you see in a completely dark room?

10 Wax paper is...

A transparent
B translucent
C opaque

11 Why does a pencil look broken in water?

12 Why does a red apple look red?

PART D

Part D · Explain (4 points)

Directions: Write a paragraph. Use science words from this unit.

13 You want to keep a cup of hot cocoa warm for a long time. Design a way to do it. Name the materials you would use, explain how heat moves, and explain why your design works.

Grade 4 Science

Student Workbook

A full year of Grade 4 science built around questions, investigations, and evidence. Students observe, measure, model, and explain the natural world, then use what they learn to design solutions to real problems.

This Student Workbook holds the daily practice pages for every lesson, plus a Weekly Check at the end of each week, the unit reviews, and the unit tests, with room to show thinking and write complete answers.

INSIDE THIS YEAR

01 The Human Body

02 Energy, Heat, and Light

IN THIS BOOK

- Warm-ups, practice, and exit tickets for every lesson
- A Weekly Check for each week of lessons
- Unit reviews and unit tests
- Space to show work and explain thinking

THE COMPLETE SET



Textbook

Workbook

You are holding this book

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Each lesson has the same number in all three books, so the textbook, workbook, and teacher pages always match.